Billing state machines and command invariants

These transitions are specifications; SQL CHECK lists alone do not enforce them. Implement authenticated command functions/services with expected state/version, row locks, idempotency, immutable evidence, audit and outbox.

Subscription

Draft -> trialing: approved trial eligibility, duration and limits.
Draft -> active: executed commercial authority, covered tenant, valid price/items, activation conditions. Payment may not be required under approved net terms.
Trialing -> active: conversion approval. Trialing -> expired: trial expiry without conversion.
Active -> past_due: issued overdue collectible obligation under policy.
Past_due -> active: allocation/approved resolution restores good standing.
Past_due -> suspended: grace exhausted and authorized suspension decision.
Suspended -> active: approved resolution and entitlement restoration.
Active/past_due/suspended -> canceled: authorized cancellation reaches effective date.
Active -> active on renewal: advance period and apply approved next-period terms once.
Period-end cancellation is a flag/scheduled event; it does not instantly set canceled.
Entitlements are effective-dated; contractual grace and continuity policies apply. Never delete education records as a billing consequence.

Invoice

Draft -> issued: invoice numbering, customer/legal/tax details, line totals, currency, approved pricing complete.
Issued -> partially_paid: eligible settled allocations below balance.
Issued/partially_paid -> paid: allocations plus applied credits cover balance.
Issued -> void only where lawful/accounting policy permits and no incompatible allocations; preserve evidence.
Issued/partially_paid -> uncollectible only after authorized write-off decision; do not claim cash received.
Issued invoices are not edited arbitrarily; approved credit records preserve original evidence.

Payment intent and payment

Created -> requires_action/processing; requires_action -> processing; processing -> succeeded/failed; cancel only while rail permits.
Intent succeeded is not automatically bank settlement. Payment pending -> settled/failed using verified rail evidence/reconciliation.
Payment allocation requires settled status, same account/currency, allocation sum <= eligible payment and invoice balances, locks on affected rows. Multi-invoice allocations follow explicit policy.
Re-query provider state for out-of-order/ambiguous events. Native inbox uniqueness scopes

provider+merchant+event. Inbox verified_at must be populated by trusted signature-verification path, not caller assertion.

Refund and credit

Refund requested -> approved -> processing -> succeeded/failed. Approval requires amount <= refundable settled balance after prior refunds/reservations; external call uses stable idempotency. Failure preserves request/evidence for reviewed retry.
Credit draft -> approved -> applied, or void before applicable posting restrictions. Credit is not a refund and may change receivable rather than cash.

Ledger

Draft -> posted only with balanced debit/credit totals per currency, valid accounts, locked source, unique posting identity. Posted entries are immutable; correction is a reversal/adjusting entry. SQL line checks do NOT enforce whole-entry balance or allocations/refund aggregate ceilings. These are mandatory command-layer controls.

API/webhooks

Commands: create_trial, activate_subscription, schedule_change, cancel_at_period_end, issue_invoice, create_payment_intent, process_verified_payment_event, allocate_settlement, approve_credit, request_refund, approve_refund, reconcile, post_journal.
Actor from session or narrow machine identity, never untrusted JSON. Persist transition and outbox transactionally. Deliver notifications asynchronously with durable retries/dead-letter. Failure of notification does not undo posted money or academic records.